

Backing up EFS file systems

To begin with the Lab

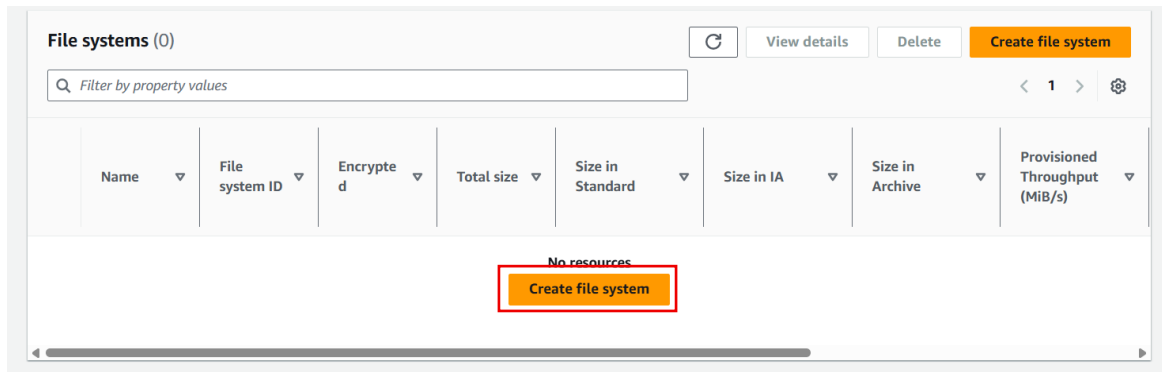
Summary of the Lab

In this lab, an Amazon EFS file system is created and backed up using AWS Backup. After enabling EFS in backup settings, an on-demand backup is initiated by selecting the EFS file system and creating a new backup vault. Once the backup completes, it is restored to a new file system using a recovery point. The restored data is verified in EFS. Additionally, automatic backup can be enabled by editing EFS settings and checking the automatic backup option.

Go to AWS Console

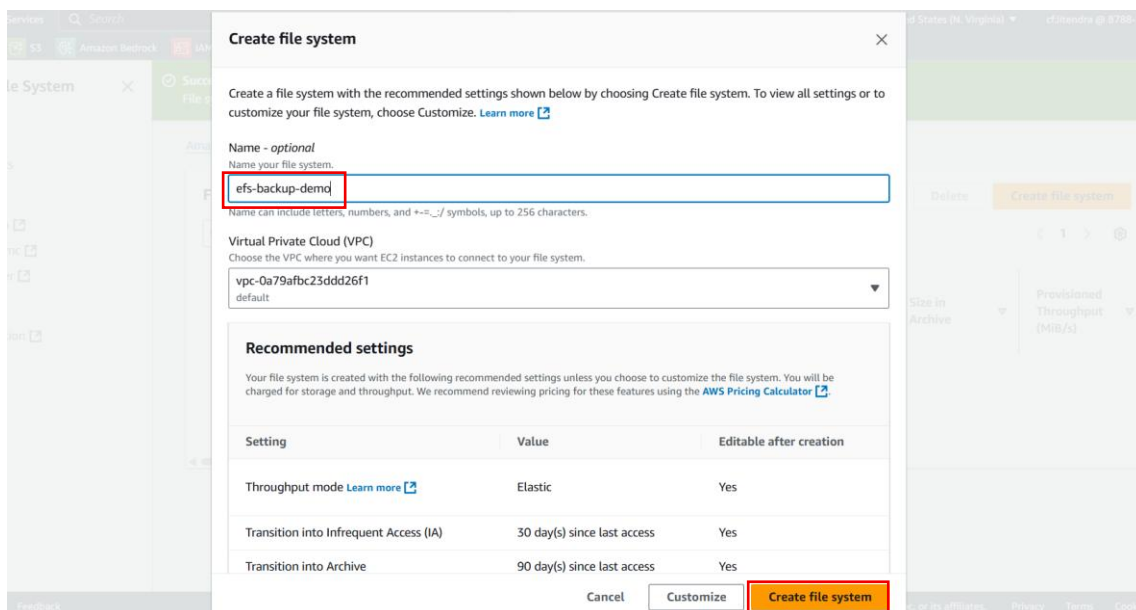
Navigate to EFS.

Click Create file system.

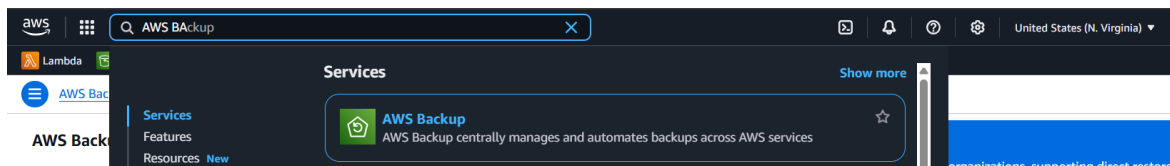


Name the EFS file

Click create.



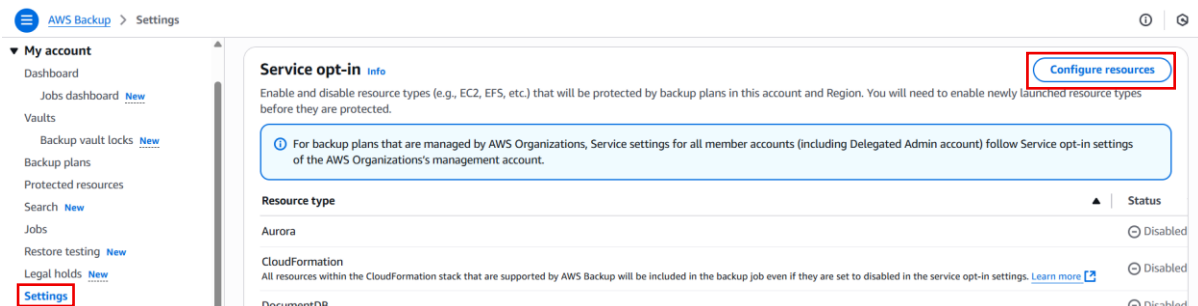
Now Navigate to AWS Backup



In AWS Backup, Left navigation bar.

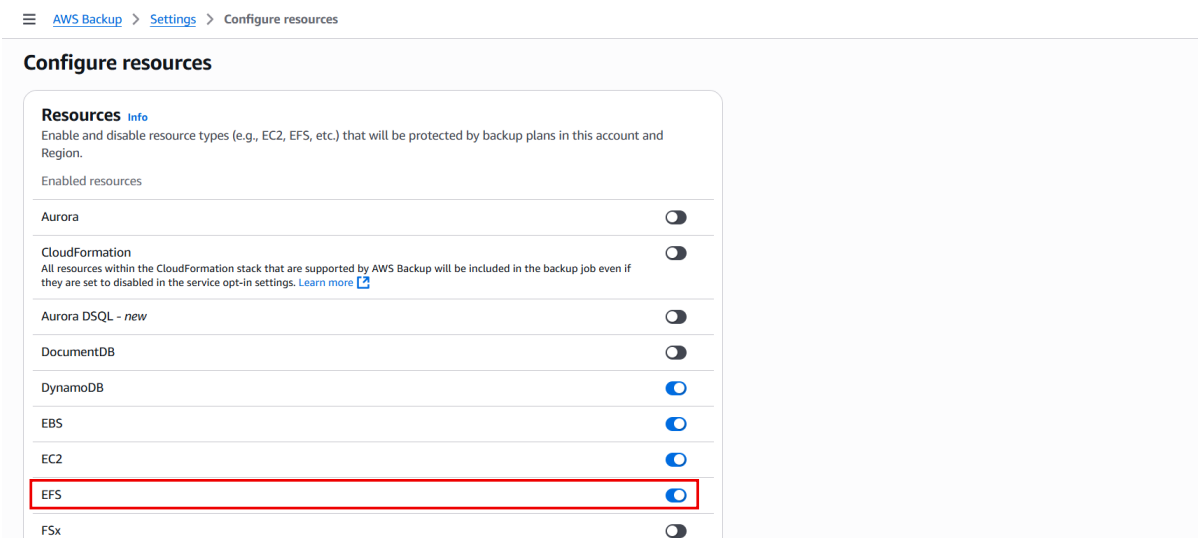
Under my account > settings.

Click on the Configure resource.



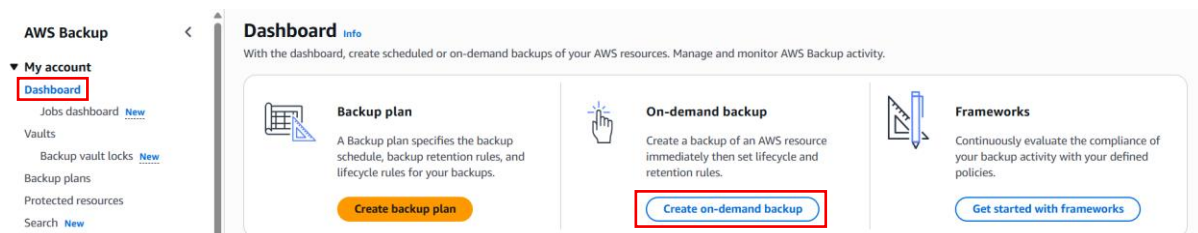
Check for the EFS configuration.

If it is off, turn it on.



Now go to Dashboard under my account.

Click on create on-demand backup



Choose EFS in the Resource Type.

And choose the file system ID.

You can select the Total Retention period according to your need.

The screenshot shows the 'Create on-demand backup' settings page. Under the 'Settings' section, 'Resource type' is set to 'EFS' and 'File system ID' is 'fs-0eb34b75ad36110e7'. In the 'Backup window' section, 'Create backup now' is selected. The 'Cold storage' section has a checkbox for 'Move backups from warm to cold storage'. The 'Total retention period' is set to '10' days.

Here I am creating a new Vault.

Click on create a new vault.

Fill the basic details.

And click create.

The screenshot shows the 'Create vault' general settings page. The 'Vault name' is 'EFS-backup-demo'. Under 'Vault type', 'Backup vault' is selected. The 'Encryption key' is set to '(default) aws/backup'.

After creating a vault.

Choose the vault.

The screenshot shows the 'Backup vault' selection dropdown. The vault 'EFS-backup-demo' is selected and highlighted with a checkmark. Other options include 'aws/efs/automatic-backup-vault'.

Click Create on-demand backup.

IAM role [Info](#)

Specify the IAM role that AWS Backup will assume when creating and managing backups on your behalf.

☒ **Default role**
If the AWS Backup default role is not present, one will be created for you with the correct permissions.

☐ Choose an IAM role

► **Tags added to recovery points - optional**
AWS Backup copies tags from the protected resource to the recovery point upon creation. You can specify additional tags to add to the recovery point.

[Cancel](#) [Create on-demand backup](#)

Your backup is in progress, once it is completed we will restore the file.

AWS Backup > Jobs

My account

- Dashboard
- Jobs dashboard [New](#)
- Vaults
- Backup vault locks [New](#)
- Backup plans
- Protected resources
- Search [New](#)
- Jobs**
- Restore testing [New](#)
- Legal holds [New](#)
- Settings

External resources

- Gateways [New](#)

Jobs

In jobs, you can monitor the status and other details of backup, restore, and copy activity.

Backup jobs | [Restore jobs](#) | [Copy jobs](#)

Backup jobs (1) [Info](#)

Records of your scheduled or on-demand backups.

[Filter backup jobs by job ID, status, resource ID, resource type or message category](#)

[Stop backup job](#) [Create report](#) [Last 24 hours](#)

Backup job ID	Status	Resource name	Message category	Resource ID	Resource type
982DF64D-4D88-37CC-323E-6D3926D7E1BA	Running	efs-backup-demo	-	file-system/fs-0eb34b75ad36110e7	EFS

Now you can see it is completed.

Backup jobs (1) [Info](#)

Records of your scheduled or on-demand backups.

[Filter backup jobs by job ID, status, resource ID, resource type or message category](#)

[Stop backup job](#) [Create report](#) [Last 24 hours](#)

Backup job ID	Status	Resource name	Message category	Resource ID	Resource type
982DF64D-4D88-37CC-323E-6D3926D7E1BA	Completed	efs-backup-demo	Success	file-system/fs-0eb34b75ad36110e7	EFS

To restore EFS backup.

Go to vaults under my account.

Select the vault.

EFS-backup-demo	Backup	-	1	c77e542d-e95e-470d-abb5-118b8f8a819a
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Select the recovery point.

Click on action.

Click on restore.

Recovery points | **Protected resources**

Recovery points (1/1) [Info](#)

Recovery point count value displayed in the console can be an approximation. See ListRecoveryPointsByBackupVault API to obtain the exact count.

[Filter by resource type, recovery point ARN, status, resource ARN or source account ID](#)

[Deselect all](#) [Actions](#)

Recovery point ID	Status	Resource name	Resource ID	Resource type
28f028d8-9d47-453c-898b-59dfd937c372	Completed	efs-backup-demo	file-system/fs-0eb34b75ad36110e7	EFS

Copy
Edit
Restore
Delete
Search backup
Create backup index

In restore location, choose restore to a new file system.

And choose the Encryption Key.

restore Backup

Settings

Backup ID
28f028d8-9d47-453c-898b-59dfd937c372

Restore type [Info](#)

☒ Full restore
Restore the filesystem in its entirety including all root level folders and files.

☐ Item-level restore
Select and restore up to 5 items within your Elastic File System. Paste or enter a relative path to a file or folder.

Restore location

☐ Restore to directory in source file system

☒ Restore to a new file system

File system type

Choose Regional to restore a file system using regional storage classes. Choose One Zone to restore a file system using One Zone storage classes.

☒ Regional - recommended
Stores data redundantly across multiple AZs

☐ One Zone
Stores data redundantly within a single AZ

Performance [Info](#)

☒ General purpose - recommended

☐ Max I/O

☒ Enable encryption
Choose to encrypt the given instance. Encryption key ids and aliases appear in the list after they have been created using the Key Management Service (KMS) console.

Encryption key [Info](#)

(default) aws/elasticfilesystem

And click on restore backup.

Restore role [Info](#)

Specify the IAM role that AWS Backup will assume when restoring the backup.

☒ Default role
If the AWS Backup default role is not present, one will be created for you with the correct permissions.

☐ Choose an IAM role

Protected resource tags [Info](#)

☒ Copy tags from the protected resource to the restored resource

[Cancel](#) [Restore backup](#)

In the restore jobs, check the status of the recovery and wait until it shows complete.

Backup jobs **Restore jobs** Copy jobs

Restore jobs (1) [Info](#) [Create report](#) [Last 24 hours](#)

Records of your backup restoration.

Filter restore jobs by job ID, status or resource ID

Restore job ID	Status	Restore time	Restore type	Resource ID	Resource type	Create time
73F73BD8-F892-7DB1-42A9-40F7AF7CCB10	Completed	1 minute	Restore	file-system/fs-07c98561ed964aa3c	EFS	July 1

To check the restore file go to EFS and check the file.

We have successfully restored the file.

File systems (2) View details Delete Create file system

Filter by property values

	Name	File system ID	Encrypt	Total size	Size in Standard	Size in IA	Size in Archive	Provisioned Throughput (MiB/s)
☐	efs-backup-demo	fs-07c98561ed964aa3c	Encrypt	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-
☐	efs-backup-demo	fs-0eb34b75ad36110e7	Encrypt	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-

There is another way to create an instant backup.

Select the efs file that you want to back.

☒	efs-backup-demo	fs-0eb34b75ad36110e7	Encrypt	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-
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Click on view details.

File systems (2) View details Delete Create file system

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☐	efs-backup-demo	fs-07c98561ed964aa3c	Encrypt	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-
☒	efs-backup-demo	fs-0eb34b75ad36110e7	Encrypt	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-

Click on edit.

Amazon EFS > File systems > fs-0eb34b75ad36110e7

efs-backup-demo (fs-0eb34b75ad36110e7) Delete Attach

General Edit

Amazon resource name (ARN)	Automatic backups
<code>arn:aws:elasticfilesystem:us-east-1:878893308172:file-</code>	Enabled

Check the enable automatic backup.

General

Automatic backups

Automatically backup your file system data with AWS Backup using recommended settings. Additional pricing applies. [Learn more](#)

☒ Enable automatic backups

Lifecycle management

Automatically save money as access patterns change by moving files into the Infrequent Access (IA) or Archive storage class. [Learn more](#)

Transition into Infrequent Access (IA)

Transition files to IA based on the time since they were last accessed in Standard storage.

30 day(s) since last access ▼

Transition into Archive

Transition files to Archive based on the time since they were last accessed in Standard storage.

90 day(s) since last access ▼

Transition into Standard

Transition files back to Standard storage based on when they are first accessed in IA or Archive storage.

None ▼

And click save changes.